

WINNER ENGINEERING INTERNAL TRAINING SERIES

MODULE III

STATIC PRESSURE CALCULATION



Winner Engineering Pte Ltd
永胜工程私人有限公司
Excellence in Service

CONTENT

1. Definition & Importance
2. Basic Calculation Approach
3. Advance Calculation

BASIC NOMENCLATURE

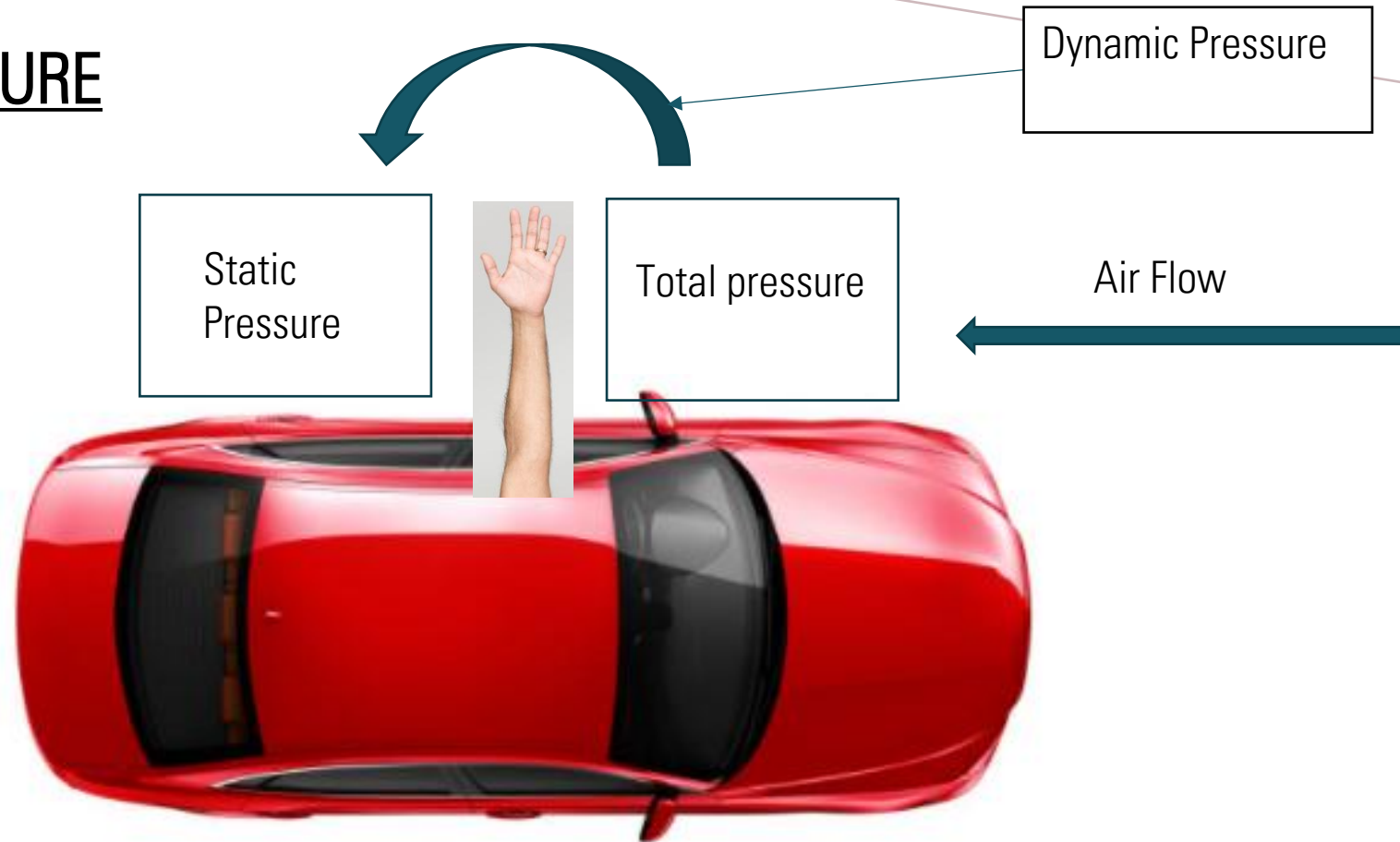
TERM	DESCRIPTION
Air Flow	Volume of air produced by fan measured by time.
Pascal	SI derivative unit for pressure
Static Pressure	Amount of air pressure that can be produced by the fan in an enclosure. It can be positive or negative. Independent of air velocity
Dynamic Pressure	Pressure created with the movement of air. Dependent of air velocity
Total Pressure	Sum of static and dynamic pressure

STATIC PRESSURE

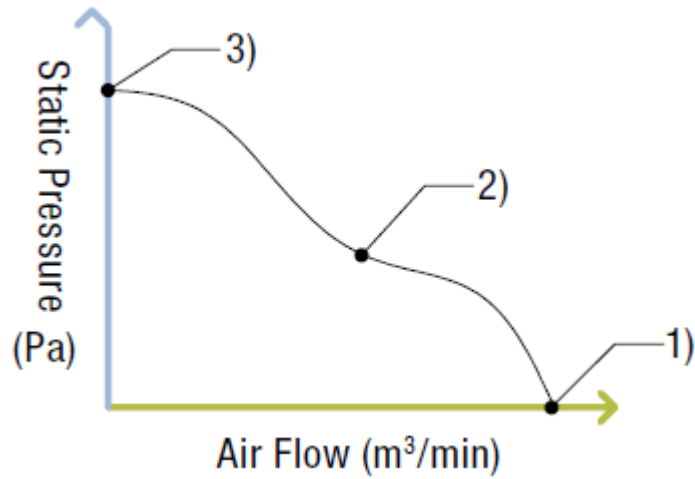


Static pressure is the pressure of a fluid or gas if it was not moving. An example would be the pressure inside a balloon. It is used to determine a fans power by showing how much it would increase the static pressure if a fan was blowing into a sealed system. Imagine using a fan to inflate a very weak balloon, a fan with a higher static pressure would inflate the balloon to a much larger size than one with a lower static pressure. The static pressure therefor can indicate how strong a fan is and how good it is at overcoming resistance.

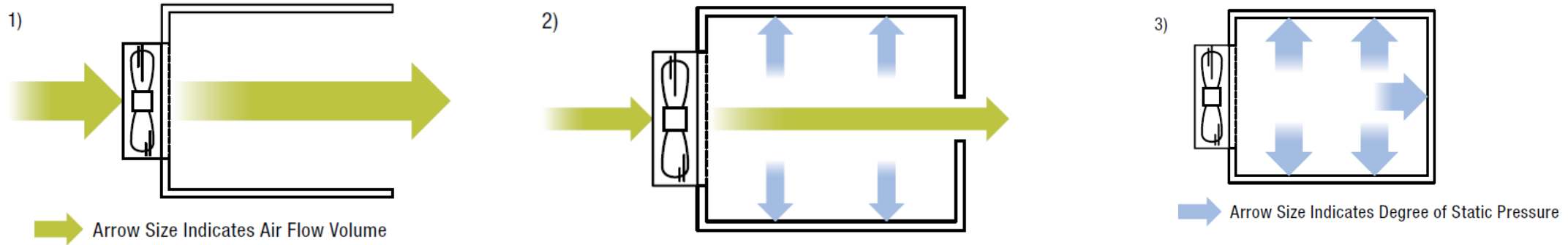
DYNAMIC PRESSURE



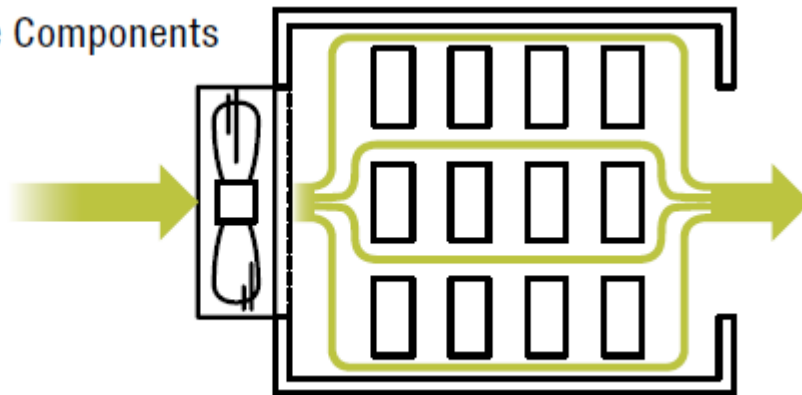
As the name suggests this is pressure created by the movement of air, it increases as velocity increases and is always positive. An example of this can be shown by sticking your hand out of a car window when it is moving. Your hand will be pushed backwards because the pressure in front of your hand is higher than the pressure behind your hand. This is because your hand is bring the air to rest and providing a pressure difference equal to the dynamic pressure of the fluid/moving air. The faster the car is going, the more force pushing your hand back, this is because the dynamic pressure of the air is higher. Note that driving faster does not increase the static pressure at all, as the air would still have the same pressure if you were standing still.



The relationship between air flow and static pressure of a fan is shown in the graph. As you can see, air flow and static pressure have a negative correlation. When air flow increases, static pressure decreases; and when static pressure increases, air flow decreases. The 3 points depict possible scenarios where the fan will perform.



With More Components



With more components (high installation density), there are more obstructions in the path of the air flow. This scenario would be similar to example 2) above, which is the most common. In this case, the high static pressure could decrease the air flow to below its maximum value.

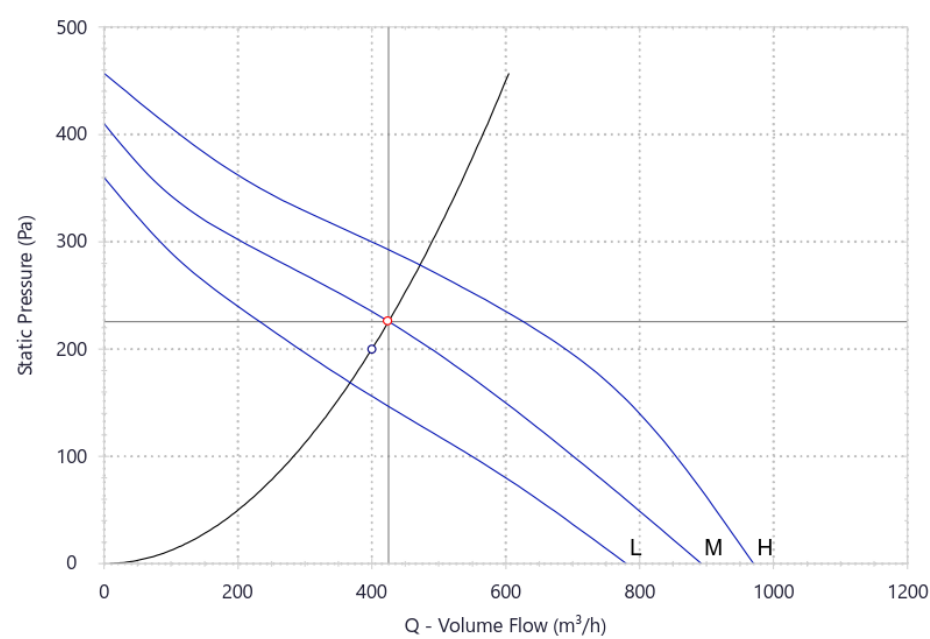
IMPORTANCE OF STATIC PRESSURE

Understanding the static pressure is pivotal when selecting the correct fan for the application. Fan selection is typically based on CMH and static pressure.

Performance

Requested Duty	400 m ³ /h@200 Pa
Actual Duty	425 m ³ /h@226 Pa
Pd (Dynamic Pressure)	8.61 Pa
Outlet Velocity	3.76 m/s
Fan Static Efficiency	19.7 %
Fan Total Efficiency	20.5 %
Absorbed Power	0.14 kW
Specific Fan Power	1.43 W/l/s
Fan Speed	2323 rpm
Sound Level	38.7 dBA
Air Density	15 °C @ 0 m 1.22 kg/m ³
Enclosure	44
Insulation	F
Motor Rating	0.14 kW

Performance Curve



IMPORTANCE OF STATIC PRESSURE

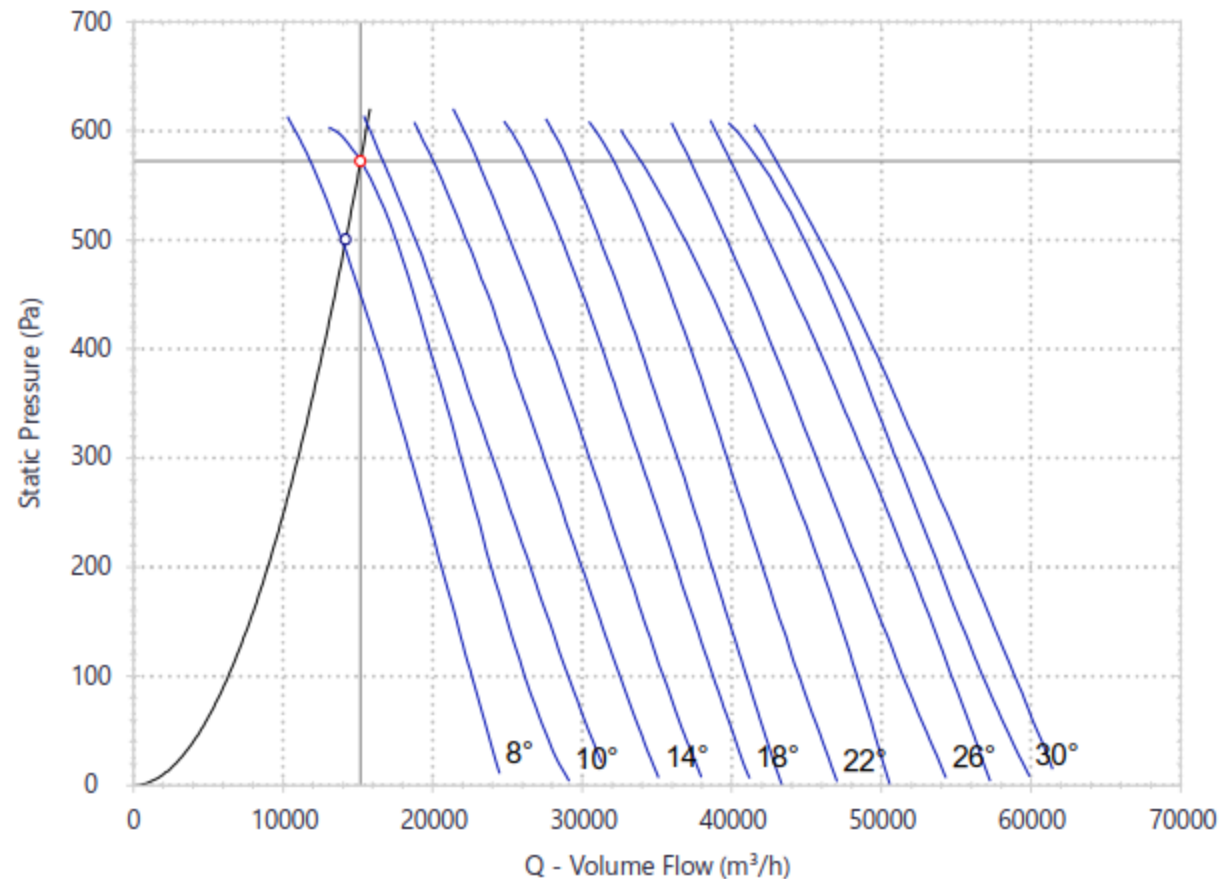
Performance

Q (Volume)	14,200 (15,193) m ³ /h
Ps (Static Pressure)	500 (572) Pa
Pt (Total Pressure)	599 Pa
Pd (Dynamic Pressure)	26.9 Pa
Outlet Velocity	6.63 m/s
Fan Static Efficiency	61.3 %
Fan Total Efficiency	64.2 %
Absorbed Power	3.94 kW
Specific Fan Power	1.17 W/l/s
Fan Speed	1440 rpm
Sound Pressure Level	80 dBA
Air Density	15 °C @ 0 m 1.22 kg/m ³
Total Weight	172Kg

Recommended Motor

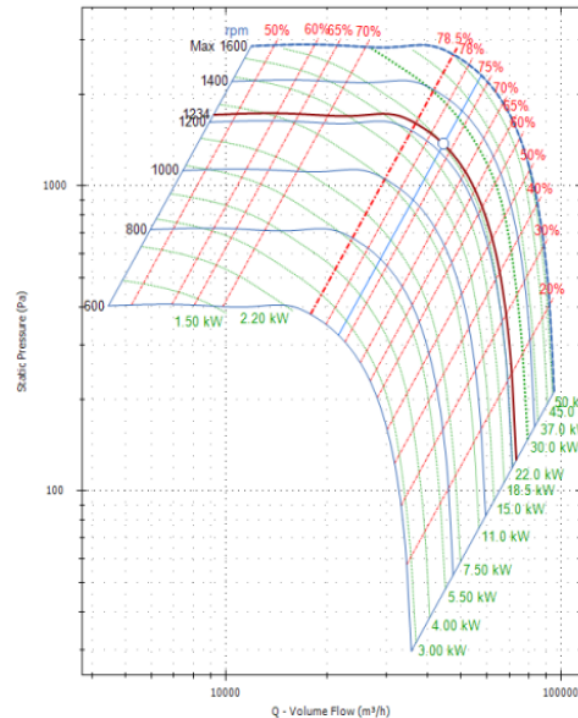
Brand	ATT-CT
Motor Speed	1440 rpm
Motor Rating	4 kW
Full Load Current	8.4 A

Performance Curve



TOTAL PRESSURE, STATIC PRESSURE, DYNAMIC PRESSURE INTERNAL STATIC PRESSURE, EXTERNAL STATIC PRESSURE

Fan Model:	BDB800X
Air Flow(CMH):	44560
Total Pressure(Pa):	1462
Static Pressure(Pa):	1372
Dynamic Pressure(Pa):	90
Outlet Vel(m/s):	12.21
Total Pre Eff:	81.4
Static Pre Eff:	76.4
Fan Speed(r/min):	1234
Absorbed Power:	22.23
Motor Power:	30
Motor Speed:	
Power Frequency:	50
Fan Pulley/Bush/Shaft Dia.:	
Motor Pulley/Bush/Shaft Dia.:	
Belt Model * Qty:	
Belt Length:	
Belt Speed:	
Max Fan Speed(r/min):	1600
Max Abs Power(kw):	50
Min Temp(°C):	-20
Max Temp(°C):	85
Bearing Breakdown Torque(n)	

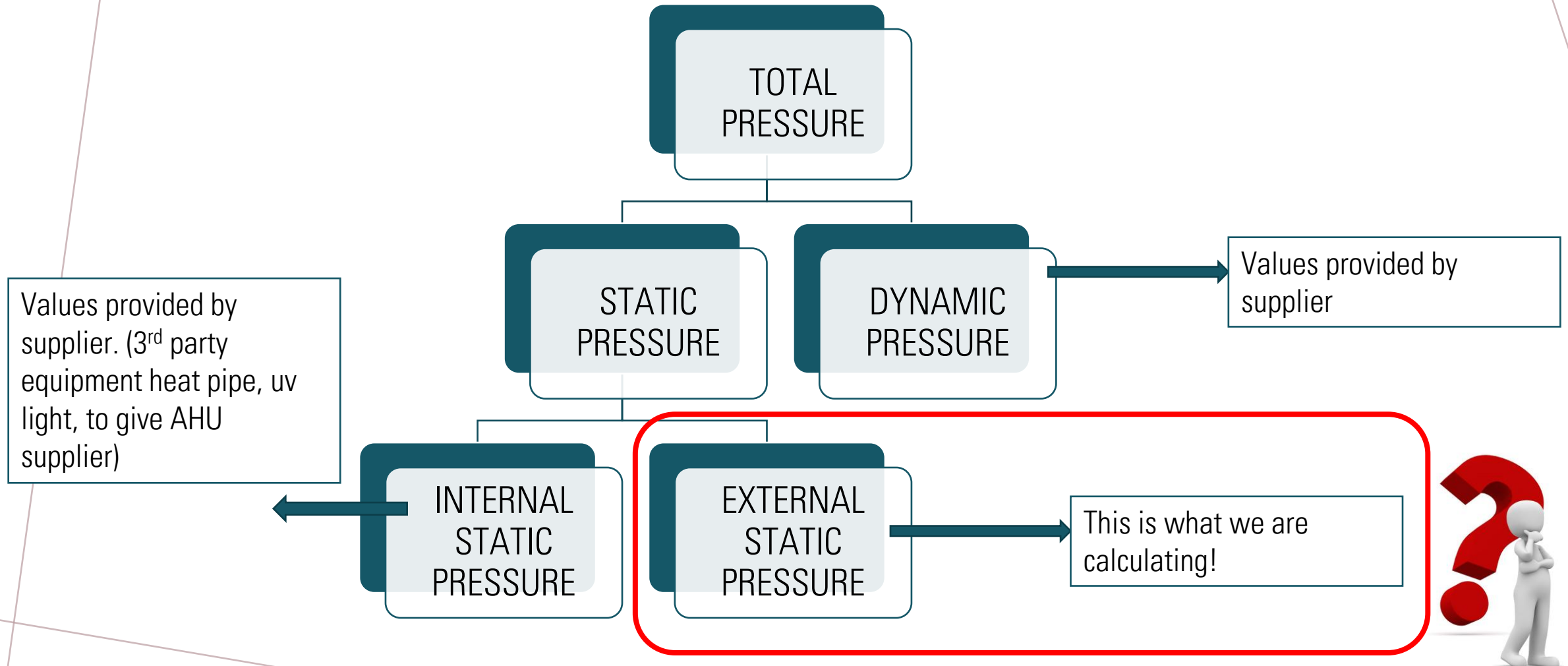


In the context of AHU, pressure drop across internally due to the coils, filter, and 3rd party add on equipment (e.g uv light, heat pipe) are considered as internal static pressure. These values are normally included in the calculation by supplier.

Essentially what we are calculating is for the external portion only.



TOTAL PRESSURE, STATIC PRESSURE, DYNAMIC PRESSURE
INTERNAL STATIC PRESSURE, EXTERNAL STATIC PRESSURE

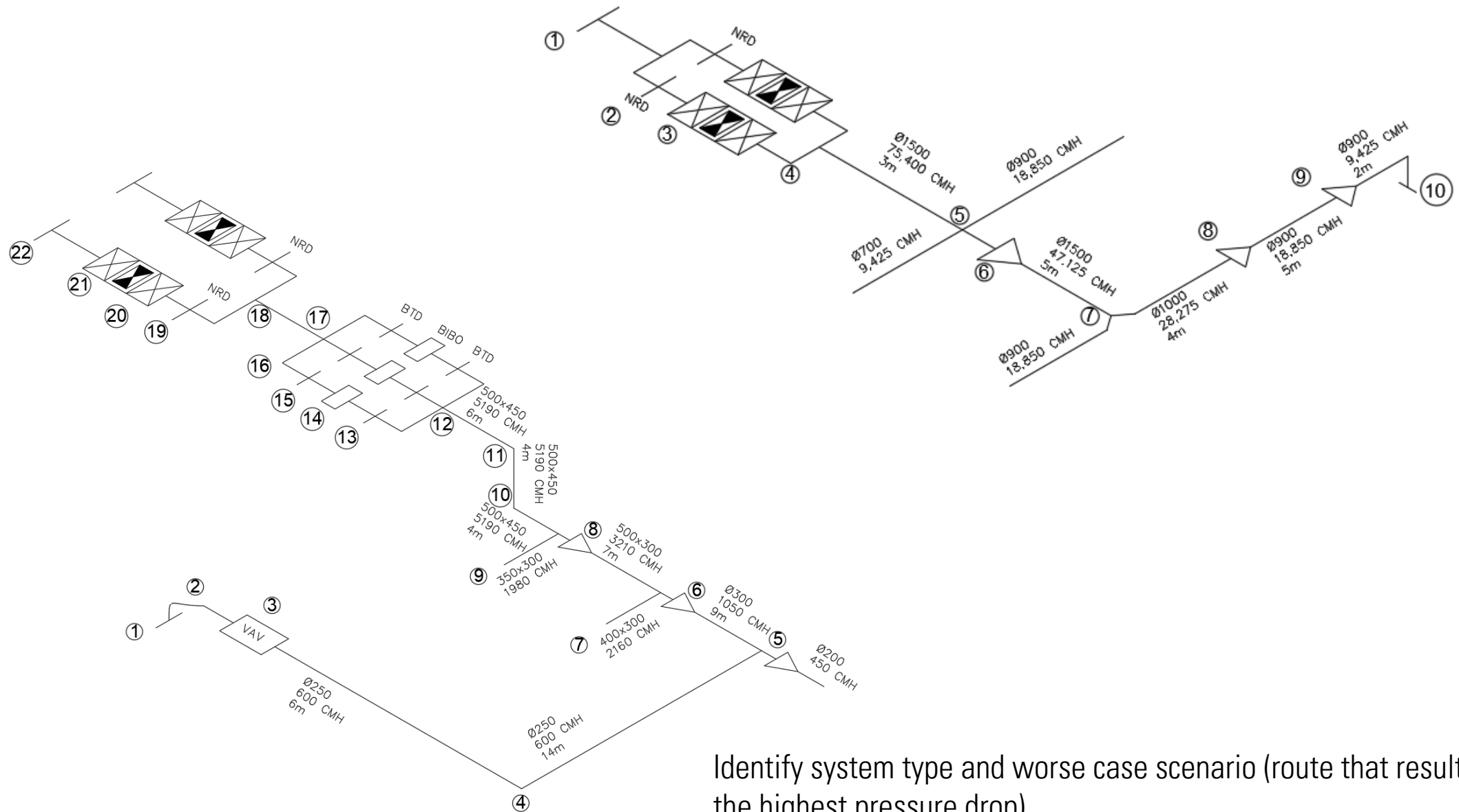


NORMAL (ESTIMATED) CALCULATION APPROACH

Step 1 : Identifying & numbering of sections & points of pressure drop

Step 2 : Preparing Airflow schematic drawing

Step 3 : Tabulate pressure drop by inputting common estimated pressure value per item



Identify system type and worse case scenario (route that resulted in the highest pressure drop)

COMMON ESTIMATED PRESSURE DROP FOR REFERENCE

ITEM	PRESSURE DROP (Pa)
Duct	1Pa/m*
Air Grille	25
Air Louvre	25
Fire Damper	10
Volume Control Damper	25
Motorised Damper	25
Non-return Damper	50
Air Tight Damper	50
Flexible	10
Duct Heater	10
Variable Air volume	150
Venturi Valve	150
Elbow	15
Tee	15

DISCLAIMER:

INDICATED VALUES ARE APPLICABLE TO LOW TO AVERAGE CMH RANGE

HIGH VOLUME FANS ARE ADVISABLE TO REFER TO MANUFACTURER CATALOGUE FOR CRITICAL ITEM. (NRD, MERV 14 FILTER, BIBO FILTER, HEPA FILTER,)

COMMON ESTIMATED PRESSURE DROP FOR REFERENCE

ITEM	PRESSURE DROP (Pa)
Fume Hood	100
SILENCER (NON-PODDED)	30
Duct Heater	20
AHU Coil	150
Pre Filter	50
Secondary Filter	250
BIBO Filter	350-400
Electronic Air Cleaner	100

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NORMAL (ESTIMATED) CALCULATION APPROACH

PROJECT:

SUBJECT: AIR CONDITIONING AND MECHANICAL VENTILATION
INSTALLATION SYSTEM - STATIC PRESSURE CALCULATION SHEET.

SCHEMATIC DRAWING : AIR FLOW CMH

CALCULATION

ITEM REF.	DESCRIPTION	QTY.	FRICTION LOSS PER UNIT (Pa/unit)	TOTAL FRICTION LOSS (Pa)	REMARKS
1					
2					
3					
4					
5					
6					
7					
8					
9					
			SUBTOTAL		
			ALLOW 20% SAFETY FACTOR		
			TOTAL		

ADVANCED CALCULATION APPROACH

Step 1 : Identifying & numbering of sections & points of pressure drop

Step 2 : Preparing Airflow schematic drawing

Step 3 : Tabulate pressure drop by cross-checking ASHRAE fitting loss coefficient & respective equipment catalogue

Essential side note, static calculation is very subjective. The values provided in be it from ASHRAE source, catalogue, are bound to have discrepancies in real life.

Not to mention variable external circumstances material type, air temperature, duct formation discrepancies, system operation, and site as built condition. What we can do is consideration for the worst case scenario approach.

ADVANCED CALCULATION APPROACH

Step 1 : Identifying & numbering of sections & points of pressure drop

Step 2 : Preparing Airflow schematic drawing

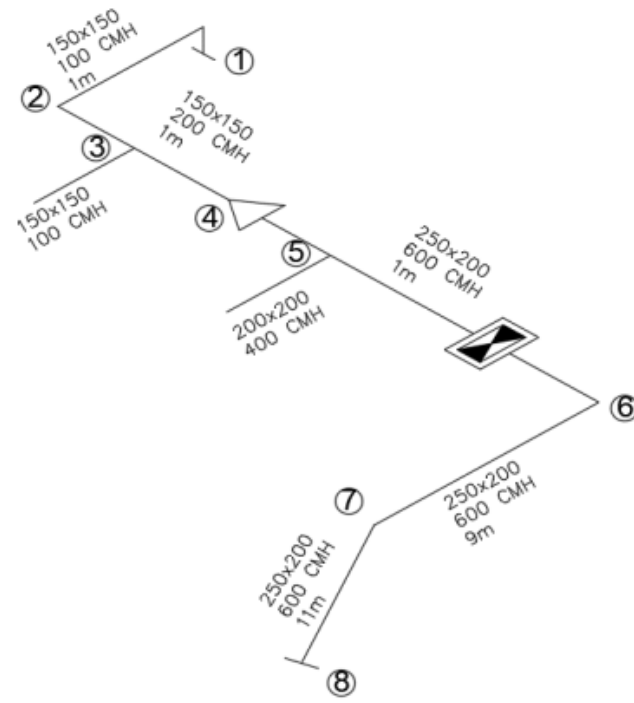
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Not to mention variable external circumstances material type, air temperature, duct formation discrepancies, system operation, and site as built condition. What we can do is consideration for the worst case scenario approach.

ADVANCED CALCULATION APPROACH

WINNER ENGINEERING (STATIC PRESSURE CALCULATION)										EAF 1-10 (600CMH)										
Fitting Number	Duct Element	ASHRAE Fitting Number	Airflow Rate	Airflow Rate	Airflow Rate	Duct size (mm)		Equivalent Dia.	Duct Area	Velocity	Velocity Pressure	Duct Length/Nos	Loss coefficients	Hydraulic Diameter	Absolute Roughness	Renold Number	Friction Factor	Duct Pressure loss (calculated)	Total Pressure Loss	
			(L/S)	(CMH)	(CMS)	Round	Rectangular	(mm)	(m²)	(m/s)	(Pa)	(m)	(Co)	Dh (mm)	(mm)	Re	f'	(Pa/m)	(Pa)	
1	EA Grille					Based on Manufactured						1							25.0	
1 to 2	Duct		28	100	0.03		150	150	164	0.023	1.24	0.93	1	-	150	0.09	12350.4	0.031	0.19	0.2
2	Elbow 90	CR3-1	28	100	0.03		150	150	164	0.023	1.24	0.93	1	0.210	150	0.09	-	-		0.2
3	Tee	ER5-3	28	100	0.03		150	150	164	0.023	1.24	0.93	1	2.230	150	0.09	-	-		2.1
3 to 4	Duct		56	200	0.06		150	150	164	0.023	2.47	3.68	1	-	150	0.09	24601.2	0.026	0.65	0.6
4	Transition	ER4-2	56	200	0.06		150	150	164	0.023	2.47	3.68	1	0.280	150	0.09	-	-		1.0
5	Tee	ER5-3	56	200	0.06		250	200	245	0.050	1.12	0.76	1	5.550	222	0.09	-	-		4.2
5 to 6	Duct		167	600	0.17		250	200	245	0.050	3.34	6.72	1	-	222	0.09	49283.6	0.022	0.68	0.7
6	Elbow 90	CR3-1	167	600	0.17		250	200	245	0.050	3.34	6.72	1	0.230	222	0.09	-	-		1.5
6 to 7	Duct		167	600	0.17		250	200	245	0.050	3.34	6.72	9	-	222	0.09	49283.6	0.022	0.68	6.1
7	Elbow 45	CR3-1	167	600	0.17		250	200	245	0.050	3.34	6.72	1	0.150	222	0.09	-	-		1.0
7 to 8	Duct		167	600	0.17		250	200	245	0.050	3.34	6.72	11	-	222	0.09	49283.6	0.022	0.68	7.5
8	EA Louvre					Based on Manufactured						1							25.0	
																		Total Static pressure	75	
																		Safety Factor (10%)	8	
																		Grand Total	83	



ADVANCED CALCULATION APPROACH

Equivalent Diameter

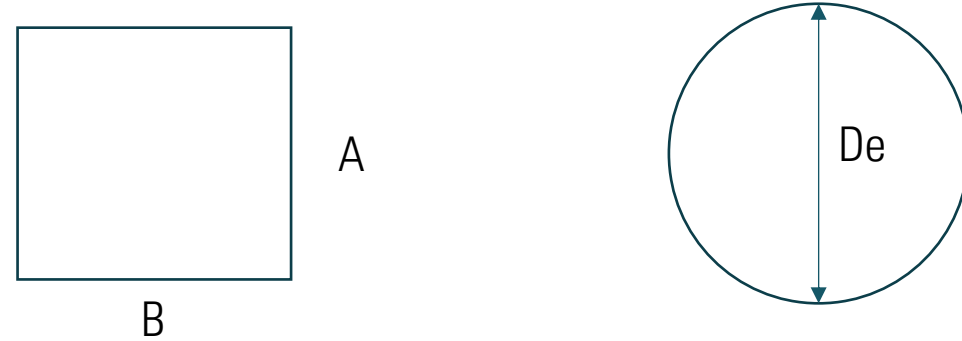
$$De = 1.3(a \times b)^{0.625} / (a+b)^{0.25}$$

Velocity

$$\text{Air Flow (cms)} / \text{Surface area(m}^2\text{)}$$

Velocity Pressure

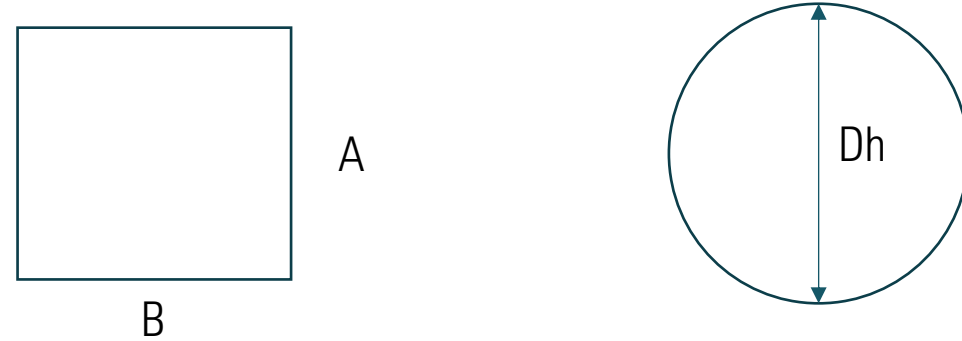
$$(\text{Air density} \times \text{Velocity}^2) / 2$$



ADVANCED CALCULATION APPROACH

Hydraulic Diameter

$$D_h = \frac{4ab}{a+b}$$



Reynold's Number

$$Re = \frac{uL}{\nu}$$

U = Air Flow CMH/3600 (m/s)

L = Characteristic linear dimension [Hydraulic diameter (mm)]

V = Kinematic viscosity (m/s) * *vary with air temperature, higher with lower temperature*

ADVANCED CALCULATION APPROACH

Friction Factor (Turbulent)

$$f = \frac{0.25}{\left[\log \left(\frac{\varepsilon/D}{3.7} + \frac{5.74}{\text{Re}^{0.9}} \right) \right]^2}$$

Swamee-Jain Equation

E = Relative roughness height (mm)

D = Air Flow CMH/3600 (m/s)

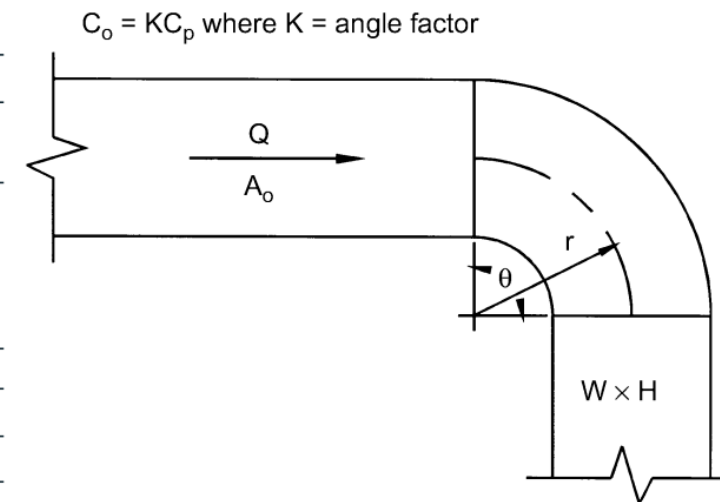
ADVANCED CALCULATION APPROACH

Identify the following in order

1. Type of System : Supply or Exhaust?
2. Type of Duct : Rectangular or Round?
3. Fittings : Elbow, transition, tee, branch, etc.
4. To refer to ASHREA database with respective parameter values involved
5. *In the event no identical reference fitting found, to refer to the most similar fitting as a guide.*

CR3-1 Elbow, Smooth Radius, Without Vanes

r/W	C_p Values										
	H/W										
	0.25	0.50	0.75	1.0	1.50	2.0	3.0	4.0	5.0	6.0	8.0
0.50	1.53	1.38	1.29	1.18	1.06	1.00	1.00	1.06	1.12	1.16	1.18
0.75	0.57	0.52	0.48	0.44	0.40	0.39	0.39	0.40	0.42	0.43	0.44
1.00	0.27	0.25	0.23	0.21	0.19	0.18	0.18	0.19	0.20	0.21	0.21
1.50	0.22	0.20	0.19	0.17	0.15	0.14	0.14	0.15	0.16	0.17	0.17
2.00	0.20	0.18	0.16	0.15	0.14	0.13	0.13	0.14	0.14	0.15	0.15
Angle Factor K											
θ	0	20	30	45	60	75	90	110	130	150	180
K	0.00	0.31	0.45	0.60	0.78	0.90	1.00	1.13	1.20	1.28	1.40

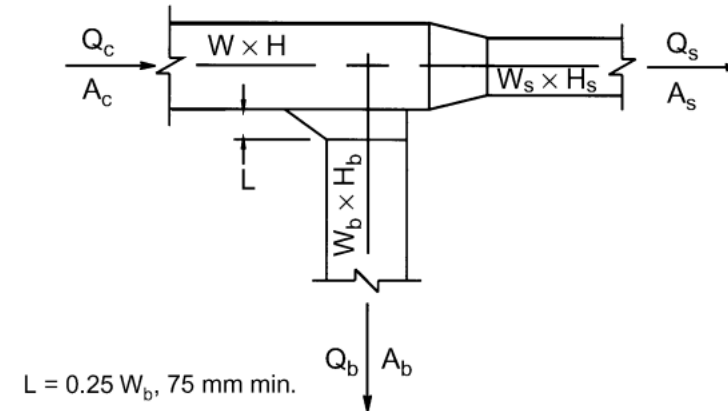


ADVANCED CALCULATION APPROACH

SR5-13 Tee, 45 Degree Entry Branch, Diverging

C_b Values									
Q_b/Q_c									
A_b/A_c	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.1	0.32	0.33	0.32	0.34	0.32	0.37	0.38	0.39	0.40
0.2	0.31	0.32	0.41	0.34	0.32	0.32	0.33	0.34	0.35
0.3	1.86	1.65	0.73	0.47	0.37	0.34	0.32	0.32	0.32
0.4	3.56	3.10	1.28	0.73	0.51	0.41	0.36	0.34	0.32
0.5	5.74	4.93	2.07	1.12	0.73	0.54	0.44	0.38	0.35
0.6	8.48	7.24	3.10	1.65	1.03	0.73	0.56	0.47	0.41
0.7	11.75	10.00	4.32	3.31	1.42	0.98	0.73	0.58	0.49
0.8	15.57	13.22	5.74	3.10	1.90	1.28	0.94	0.73	0.60
0.9	19.92	16.90	7.38	4.02	2.46	1.65	1.19	0.91	0.73

C_s Values									
Q_s/Q_c									
A_s/A_c	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.1	0.04	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.2	0.98	0.04	0.01	0.00	0.00	0.00	0.00	0.00	0.00
0.3	3.48	0.31	0.04	0.01	0.00	0.00	0.00	0.00	0.00
0.4	7.55	0.98	0.18	0.04	0.02	0.00	0.00	0.00	0.00
0.5	13.18	2.03	0.49	0.13	0.04	0.00	0.01	0.00	0.00
0.6	20.38	3.48	0.98	0.31	0.10	0.04	0.02	0.01	0.00
0.7	29.15	5.32	1.64	0.60	0.23	0.09	0.04	0.02	0.01
0.8	39.48	7.55	2.47	0.98	0.42	0.18	0.08	0.04	0.02
0.9	51.37	10.17	3.48	1.46	0.67	0.31	0.15	0.07	0.04

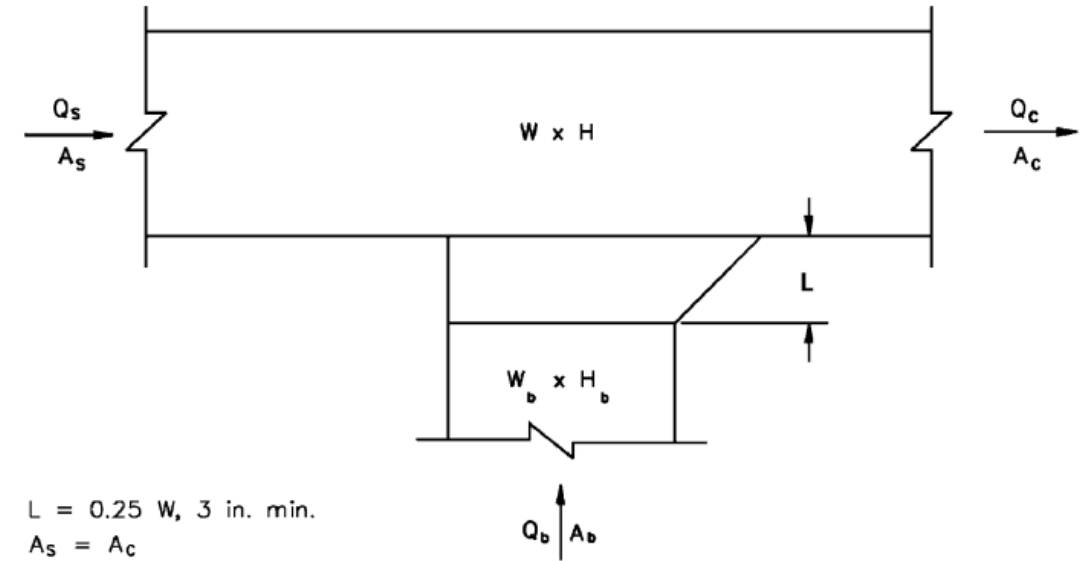


ADVANCED CALCULATION APPROACH

ER5-3 Tee, 45 Degree Entry Branch, Converging

Q_b/Q_c	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
C_b	-19.38	-3.75	-0.74	0.48	0.66	0.75	0.85	0.77	0.83	0.83

Q_s/Q_c	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
C_s	22.15	11.91	6.54	3.74	2.23	1.33	0.76	0.38	0.10	0.0



$L = 0.25 W, 3 \text{ in. min.}$

$A_s = A_c$

$A_b/A_c = 0.5$

PERFORMANCE DATA FOR GSD SERIES

CEILING DIFFUSER

600×600 FACE SIZE

SIZE (mm)	CAP l/s	50	60	70	80	90	100	120	140	160	180	200	250	300	350	400	450	500	600	700
150X150	S.P.	11	15	20	26	32	39	58	78											
	NR	14	18	22	27	30	35	40	45											
	THROW	2	3	3	3.5	4	4.5	5	5.5											
200X200	S.P.			11	14	18	22	33	45	54	73									
	NR			15	18	21	25	30	35	40	45									
	THROW			3	3.5	3.5	4	4.5	5	5.5	6									
250X250	S.P.						9	15	18	20	31	40	52	71						
	NR						15	18	21	27	31	33	40	48						
	THROW						3.5	4	4.5	5	5.5	5.5	6.5	7						
300X300	S.P.							7	9	12	17	20	27	31	54	64	90			
	NR							12	15	18	20	23	28	35	40	45	50			
	THROW							3.5	4	4.5	5	5	6	6.5	6.6	7.5	7.5			
350X350	S.P.								5	7	8	10	15	21	30	36	70	55		
	NR								11	13	15	17	21	27	31	35	39	43		
	THROW								4	4	4.5	5	5.5	6	6.5	7	7	7.5		
400X400	S.P.									6	7	8	12	17	22	28	38	43	60	
	NR									11	13	15	18	25	27	31	35	38	44	
	THROW									4.5	4.5	5	5.5	6	6.5	7	7	7.5	8	
450X450	S.P.											6	10	13	18	23	32	35	49	65
	NR											13	17	19	24	29	32	35	40	45
	THROW											5	6	6	6.5	6.5	7	7	8	8

Pre-Filtration G3 to M5 | Pleated Filters

30/30®



Advantages

- Water resistant cardboard frame
- Robust construction
- Bonded cross members to maintain pleat spacing
- Fully supported media bonded onto a wire support grid
- Radial pleat design for full media utilization
- Replaceable filter media

Application: Prevention of dust and dirt build up on heating/cooling coils within ventilation systems

Type: Prefilter

Frame: Rigid water resistant cardboard

Media : Blended polyester

Recommended final pressure drop: 250 Pa

Temperature: 70°C maximum in continuous service

Holding frames: Front and side access housings and frames are available, Type 8 and FC Housings

Fire rating: UL 900



Pre-filter

Average of initial PD and final PD
 $[(80+250)/2] = 165 \text{ Pa}$

Pressure drop across filters

$165+315 = 480 \text{ Pa}$

Hepa Filter

Average of initial PD and final PD
 $[(210+420)/2] = 315 \text{ Pa}$

Article Number	Type	ASHRAE 52.2:2017	EN779:2012	ISO16890	Dimensions WxHxD (mm)	Air Flow/Pressure drop (m³/hr/Pa)	Media area (m²)	Weight (kg)
54862001	30/30 20x16x1	MERV A 8A	G4	ePM10 50%	394 x 495 x 22	1326/60	0,5	0,3
54862002	30/30 20x20x1	MERV A 8A	G4	ePM10 50%	495 x 495 x 22	1650/60	0,61	0,4
54862003	30/30 25x20x1	MERV A 8A	G4	ePM10 50%	495 x 622 x 22	2065/60	0,74	0,55



EN 1822 (Efficiency @ MPPS): E10 (≥85%), E11 (≥95%), E12 (≥99.5%), H13 (≥99.95%), H14 (≥99.995%)

Rec. final pressure drop: 2x Initial pressure drop

Max. final pressure drop: 600 Pa

Temperature max: 70°C

RH. max: 100%

Mounting/Frames: FKB, CamBox, CamSafe 2, CamCube

Remarks: Special versions on request (e.g. stainless steel frame or high temperature version 120°C)

Art. No.	Model Name	EN1822	Dimensions WxHxD (mm)	Air Flow (m³/h)	Pressure Drop (Pa) *	Media Area (m²)	Weight (kg)
ABV3021131000	VEXL10-610x305x292-P	E10	610x305x292	1500/1800	150/190	10,9	6,0
ABV6021121000	VEXL10-610x610x292-P	E10	610x610x292	3400/4000	150/190	22,0	9,0
ABV3031131000	VEXXL10-610x305x292-P	E10	610x305x292	2300	210	14,9	6,4
ABV6031121000	VEXXL10-610x610x292-P	E10	610x610x292	5000	210	30,2	9,9
ABV3121131000	VEXL11-610x305x292-P	E11	610x305x292	1500/1800	170/210	18,9	6,8
ABV6121121000	VEXL11-610x610x292-P	E11	610x610x292	3400/4000	170/210	38,3	10,7
ABV3221131000	VEXL12-610x305x292-P	E12	610x305x292	1500/1800	180/220	18,9	6,8
ABV6221121000	VEXL12-610x610x292-P	E12	610x610x292	3400/4000	180/220	38,3	10,7
ABV3321131000	VEXL13-610x305x292-P	H13	610x305x292	1500/1800	200/240	20,5	6,9
ABV6321121000	VEXL13-610x610x292-P	H13	610x610x292	3400/4000	200/240	41,6	11,0
ABV3331131000	VEXXL13-610x305x292-P	H13	610x305x292	2300	400	20,5	6,9
ABV6331121000	VEXXL13-610x610x292-P	H13	610x610x292	5000	400	41,6	11,0
ABV3421131000	VEXL14-610x305x292-P	H14	610x305x292	1500	250	20,5	6,9
ABV6421121000	VEXL14-610x610x292-P	H14	610x610x292	3400	250	41,6	11,0
ABV3431131000	VEXXL14-610x305x292-P	H14	610x305x292	1800	310	20,5	6,9
ABV6431121000	VEXXL14-610x610x292-P	H14	610x610x292	4000	310	41,6	11,0

* Pressure drop ± 10 %

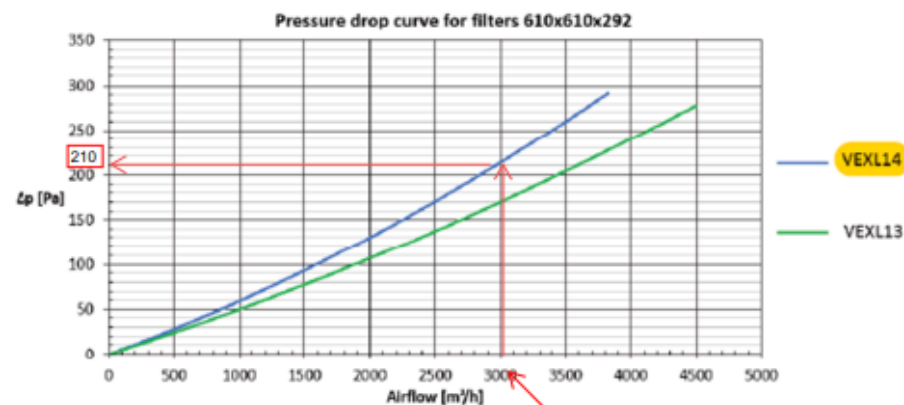


TABLE 1.1 for AIR FILTERS and EFFICIENCY

CLASSIFICATION	Arrestance or Dust Spot Efficiency	US ASHRAE 52.2	European Union EN779 Class		Typical Controlled Contaminant	Application	MAYAIR Range of Products
PRE Filter (G Class)	AFI <65 %	MERV 1	G1	Am< 65 %	Particle bigger than 10.0µm (Pollen) (Spanish moss) (Dust mites) (Sanding dust) (Spray paint dust) (Textile fibers)	Gross filter, domestic and commercial	M-NET
	AFI 65 %-70 %	MERV 2	G2	65 % ≦ Am< 80 %			M-WASH
	AFI 70 %-75 %	MERV 3					ROLL MEDIA
	AFI 75 %-80 %	MERV 4					
	AFI 80 %-85 %	MERV 5	G3	80 % ≦ Am<90 %	Particle size within 3.0µm-10.0µm (Mold) (Spores) (Hair spray) (Cement dust) (Snuff) (Powdered milk)	Commercial, industrial, paint shop	M-NET
	AFI 85 %-90 %	MERV 6					M-WASH
	NBS 25 %-30 %	MERV 7	G4	90 % ≦ Am			M-DP
	NBS 30 %-35 %	MERV 8					ROLL MEDIA
MEDIUM Filter (F Class)	NBS 40 %-45 %	MERV 9	F5	40 % ≦ Em< 60 %	Particle Size within 1.0µm-3.0µm (Lead dust) (Milled flour) (Coal dust) (Auto emissions) (Nebulizer drop) (Welding fumes)	IAQ concerned commercial & industrial, medical	M-PACK
	NBS 50 %-55 %	MERV 10					M-BOX
	NBS 60 %-65 %	MERV 11	F6	60 % ≦ Em< 80 %			M-MI, M-MII
	NBS 70 %-75 %	MERV 12					M-MV
	NBS 80 %-85 %	MERV 13	F7	80 % ≦ Em< 90 %	Particle size within 0.3µm-1.0µm (All bacteria) (cooking oil) (Most smoke) (Copier toner) (Most face powder) (Most paint pigments)	IAQ concerned commercial, industrial, medical, food etc	M-PACK
	NBS 90 %-95 %	MERV 14	F8	90 % ≦ Em< 95 %			M-BOX
	NBS>95 %	MERV 15	F9	95 % ≦ Em			M-MI, M-MII
		MERV 16					M-MV

TABLE 1.2 for AIR FILTERS and EFFICIENCY

CLASSIFICATION	Mean Fractional Efficiency	IENT RP-CC001.3	European Union EN1822 Class		Typical Controlled Contaminant	Application	MAYAIR Range of Products
HEPA Filter (H Class)	$\geq 95\%$ at $0.3\mu\text{m}$	n/a	H10	$\geq 85\%$ at MPPS	Particle size bigger than $0.3\mu\text{m}$ (Virus [unattached]) (Carbon dust) (Sea salt) (All combustion smoke) (Radon progeny)	All types of cleanrooms	M-HI
	$\geq 98\%$ at $0.3\mu\text{m}$		H11	$\geq 95\%$ at MPPS			M-HV
	$\geq 99.97\%$ at $0.3\mu\text{m}$	TYPE A					M-CFCN
	$\geq 99.99\%$ at $0.3\mu\text{m}$	TYPE C	H12	$\geq 99.5\%$ at MPPS			M-CAM
	$\geq 99.995\%$ at $0.3\mu\text{m}$		H13	$\geq 99.95\%$ at MPPS			M-TMI
	$\geq 99.999\%$ at $0.3\mu\text{m}$	TYPE D	H14	$\geq 99.995\%$ at MPPS			M-FFU
ULPA Filter (U Class)	$\geq 99.9995\%$ at $0.12\mu\text{m}$	TYPE F	U15	$\geq 99.9995\%$ at MPPS	Particle size bigger than $0.12\mu\text{m}$	super cleanroom	M-HV & M-CAM
	$\geq 99.99995\%$ at $0.12\mu\text{m}$		U16	$\geq 99.99995\%$ at MPPS			M-CFCN
	$\geq 99.999995\%$ at $0.12\mu\text{m}$		U17	$\geq 99.999995\%$ at MPPS			M-FFU

Note :

1. AFI : American Filter Institute
2. NBS : National Bureau of Standards
3. ASHRAE : American Society of Heating Refrigerating & Air-conditioning Engineers
4. MERV : Minimum Efficiency Reporting Value
5. MPPS : Most Penetrating Particle Size
6. HEPA : High Efficiency Particulate Air Filter
7. ULPA : Ultra Low Penetration Air Filter
8. Am : Average Arrestance Efficiency for Coarse Filters
9. Em : Average Efficiency for Fine Filters
10. IEST : Institute of Environmental Sciences and Technology

ADVANCED CALCULATION APPROACH (ALTERNATIVE)

Refer to Excel Document “Winner ESP Calculation”

“\\192.168.1.187\Design & Drafting\7. Detail & Standards\10) Winner ESP Calculation Form & Data\b)
Advance